Restaurant App - Software Requirements Specification v2

1. Purpose and scope

The app supports customers browsing a restaurant menu, placing and tracking orders, book ing tables, and submitting reviews. Managers administer menu items, orders, reservations , accounts, and dashboard summaries.

2. Actors

Customer: registers or signs in, browses the menu, creates orders and reservations, subm its reviews, and views their profile and reservations.
Manager: signs in, manages menu items, order status, reservations, customer accounts, an d dashboard information.

3. Functional requirements

FR-01 Users can register and authenticate with email and password; passwords are hashed.
FR-02 Users can browse menu items; managers can create, edit, and delete them.
FR-03 Authenticated users can place orders. Server totals are calculated from current st ored menu prices.
FR-04 Managers can update order status among pending, preparing, ready, served, and canc elled. Only pending orders can be cancelled.
FR-05 Customers can create reservations and view their own; managers can list and accept or decline reservations.
FR-06 Customers may cancel their own pending reservations.
FR-07 Signed-in customers can submit a 1-5 rating with optional text.
FR-08 Managers can view dashboard summaries and manage customer accounts.
FR-09 Protected APIs authenticate with bearer tokens and enforce role/ownership permissi ons.

4. Non-functional requirements

NFR-01 The mobile client communicates with the backend using JSON over HTTP or HTTPS.
NFR-02 Database connection strings and JWT secrets stay in server-side environment varia bles.
NFR-03 Errors use meaningful HTTP status codes and JSON error messages.
NFR-04 Mobile connectivity supports simulator, emulator, and LAN device base URLs.
NFR-05 Backend code separates routes, controllers, models, config, and middleware.

5. Data

Users have name, unique email, password hash, and role. Menu items have name, category, price, description, image, and availability. Orders keep customer, item snapshots, total s, status, and notes. Reservations reference a customer and table with time, party size, and status. Reviews reference a customer with a rating and text.

6. Interfaces

REST endpoint table is in the repository README. A Postman collection is included in thi
s folder. Architecture and data model diagrams are in diagrams/.

7. Constraints and assumptions

MongoDB is required for persistent storage. Seed accounts and menu data are for demonstr
ation. Set EXPO_PUBLIC_API_URL to a host reachable from the target device. Reviews requi
re authentication.